AI Project Proposal
<Dr.House AI>

01286342 Artificial Intelligence
Software Engineering Program

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<Dr.House AI>

Dr.House AI is similar to a demo/prototype/simple version of what robots and artificial intelligence do for diagnosing patients faster by assisting professional health care. Dr.House AI is inspired by the medical drama show “House M.D.”, Dr.House is a brilliant doctor who diagnoses patients with strange and bizarre symptoms and comes up with different cures. While this application does not come up with unique cures, we aim to make diagnosing patients as easy as the brilliant Dr.House.

Dr.House AI key features includes:

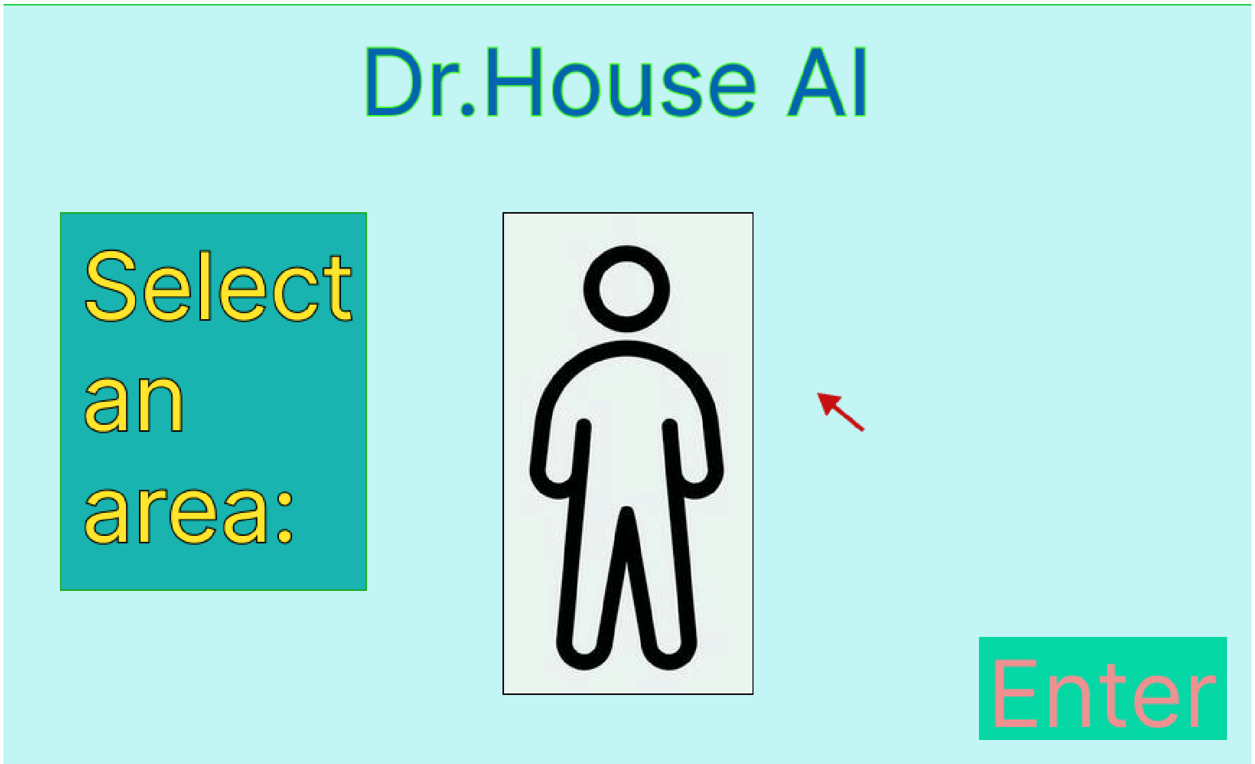
a) Symptom Input and Analysis: User inputs their symptoms through our Python GUI. The system then employs logical reasoning to analyze the symptoms and relate it to diseases with said symptoms described by users. Each symptom will have a severity measurement to tell the system how severe a symptom is and analyzes what cure to suggest.
b) Knowledge Base: The system is built upon a medical knowledge base gathered by us including various information on various diseases, criteria, and treatment options. For vague symptoms, we deploy rules to help clarify and classify those symptoms.
c) Diagnostic Recommendation: Using a rule-based approach, the system analyzes the inputted symptoms and correlates them with potential medical conditions from the knowledge base. The system then outputs a list of possible medical conditions for professional health care to investigate further.
d) Treatment suggestion: Once a diagnosis is established, the system can recommend treatment options based on modern and best practices according to the medical guidelines. This provides medication information, therapy, and lifestyle changes according to the patient’s condition.

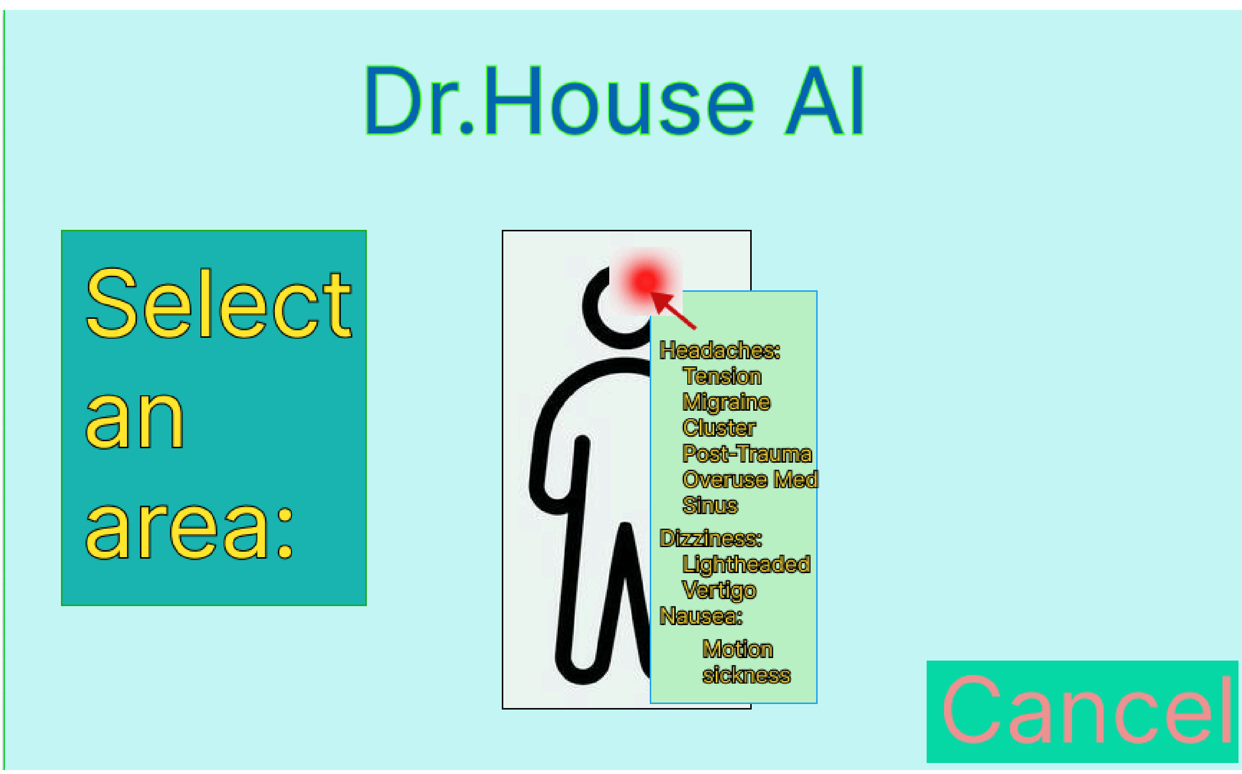
Intelligent Agent: This agent is a computer program that simulates the judgment and behavior of a human or a healthcare organization. The Dr.House AI is planned to use a rule-based approach to diagnose medical conditions. It incorporates a knowledge base of medical information, including symptoms and associated diseases. Applying logical rules to the knowledge base, the system can draw conclusions about potential diagnoses and treatment options based on user' inputs. We use the concept of what is considered an intelligent agent and using it as a framework to build upon our system, identifying the performance measure: how accurate the diagnosis is, how easy is it for users without medical knowledge to use, etc. environment: the input Python GUI, actuators: the output screen, sensors: keyboard and mouse.

Knowledge Base: Dr.House AI compiles information of diseases and symptoms from trustworthy sources then inputted into the knowledge base, queries are then made through the Python Interface to query for the diseases according to the inputted symptoms.

While at the moment our system doesn't employ a lot of concepts, more concepts might appear into our system while implementing the system. To achieve better performance, methods like searches may be employed to make the system faster while traversing the bigger knowledge base. Right now our concern is to accurately give medical guidance and diagnosis through having a reliable knowledge base.

Screen Designs:



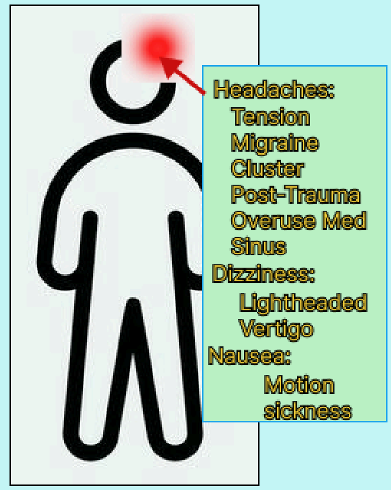


Dr.House AI

Treatments

Tension headaches primary caused by stress and muscle tension, or anxiety.

Relaxation, Physical Therapy, Cold Compression, Hydration



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